LECTURE 03 · STUDY NOTES

Risk and Return

How to measure risk and return for a single asset and for a portfolio, and why correlation is the foundation of Modern Portfolio Theory.

How to use these notes

These notes follow the order of the Lecture 03 slides, but they explain each idea in full sentences rather than slide fragments. Every numerical exercise from class is worked out step by step, with each intermediate number shown so you can check your own work line by line. A few extra examples fill in calculations the slides only state as formulas. Practice problems with full solutions are at the end. Readings: Bodie, Kane and Marcus, Chapters 5.4 and 6.1.

1. Opening: long and inverse index ETFs in 2022 and 2023

The student presentation compared $10,000 invested in QQQ, which tracks the NASDAQ-100, with $10,000 in SQQQ, which targets −3× the index's *daily* return, across the 2022 bear market and the 2023 bull market. The pattern is what you would expect in direction: QQQ fell sharply in 2022 while SQQQ rose, and in 2023 the roles reversed as QQQ rallied and SQQQ lost most of its value.

The size of the moves is the lesson. SQQQ's result over a full year is *not* −3 times QQQ's annual return, because SQQQ resets its leverage every night. On any single day the relationship is exact: if QQQ rises 2%, SQQQ falls about 3 × 2% = 6%. Over many days, compounding makes the gap between SQQQ and "−3× the index" grow with volatility. This is the volatility decay covered in the Lecture 02 notes (Examples 6 and 7). The presentation then asked a more basic question: SQQQ is a packaged bet against the index, so how does the underlying bet, a **short sale**, actually work?

2. Short selling: the mechanism behind inverse funds

Short selling means selling shares you do not own. You borrow them through your broker, sell them at today's price, and later buy them back ("cover") to return to the lender. If the price has fallen in between, you keep the difference; if it has risen, you pay it. Three groups use short sales: **speculators** who expect a price to fall, **hedge funds** that short one position to offset the risk of a long position, and **arbitrageurs** who short an overpriced security while buying a related underpriced one.

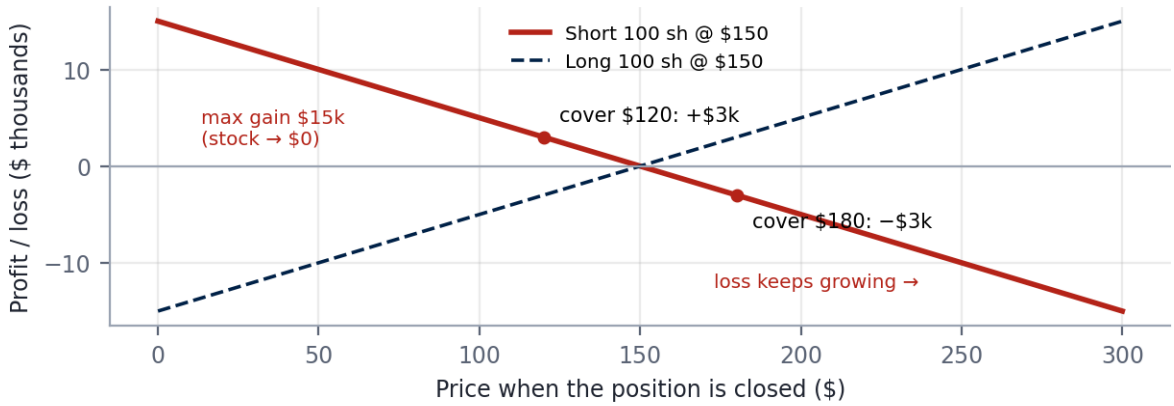
The process has five steps: borrow the shares, sell them at the market price P_0, wait, cover by buying them back at P_1, and return them. Per share, before costs:

$$\text{Short-sale profit per share} = P_0 - P_1 \text{ (sell high first, buy low later)}$$

WORKED EXAMPLE 1

Shorting 100 shares at $150

Step	What we compute	Result
1	Sell 100 borrowed shares at P_0 = \$150. Proceeds = 100 × \$150	**\$15,000**
2 · Gain	Price falls, cover at P_1 = \$120. Profit per share = \$150 − \$120	**+\$30**
	Total = \$30 × 100 shares	**+\$3,000**
	Return relative to the short-sale price = \$30 ÷ \$150	**+20.0%**
3 · Loss	Price rises, cover at P_1 = \$180. Profit per share = \$150 − \$180	**−\$30**
	Total = −\$30 × 100 shares	**−\$3,000**
	Return = −\$30 ÷ \$150	**−20.0%**
4 · Limits	Best case: stock goes to \$0 → profit = \$150 × 100	**Max gain \$15,000 (100%)**
	Worst case: no upper limit on P_1	**Loss is unlimited**



Profit or loss at closing, short versus long, 100 shares at \$150. The short's gain is capped at \$15,000; its loss has no floor. (Borrow fees and dividends are ignored here; see Lecture 02, Example 5.)

This asymmetry is the key risk of shorting. A long position can lose at most 100% of what you paid, because a price cannot fall below zero. A short position can lose more than 100% of the sale proceeds because a price can keep rising. In practice the broker will not wait: as the price rises, the equity in your margin account shrinks, and a margin call can force you to cover before you would choose to.

SQQQ: short selling, packaged

SQQQ gives investors a short position on the whole NASDAQ-100 without any of the machinery of a short sale. Instead of borrowing shares of NVIDIA, Apple, Microsoft, Amazon, Tesla and the other 95 index members, the fund enters swap contracts that pay it

the negative of the index's return. Investors buy SQQQ like any ETF, with no margin account, no borrow fee and no margin calls, and they can never lose more than they paid.

	Manual short sale	SQQQ (ProShares)
How it works	Borrow shares, sell, buy back later	Swap contracts on the NASDAQ-100
Payoff	−1 × the stock's return over the holding period	−3 × the index's *daily* return
Account needed	Margin account	Any brokerage account
Costs and risks	Borrow fees, dividends owed, margin calls, unlimited loss	Expense ratio and volatility decay; loss capped at amount invested
Beta / correlation with QQQ	n/a (single stock)	Beta −3.00; daily correlation −1.00

The beta of −3 and correlation of −1 describe the fund's *daily* behavior, which is exactly what it is built to deliver. They are the first hint of today's main topic: how the correlation between two assets determines the risk of holding them together.

3. Part 1: Return and risk for a single asset

The **holding-period return** (HPR) measures what an investment actually earned over the time it was held. It has two sources: the change in price (the capital gain or loss) and any cash income received, such as dividends:

$$\text{HPR} = (P_1 - P_0 + D) \div P_0 = \text{capital-gain yield} + \text{dividend yield}$$

WORKED EXAMPLE 2

Holding-period return

You buy a stock at $50, receive a $1.50 dividend during the year, and sell at $54. (Illustrative numbers.)

Step	What we compute	Result
1	Capital gain = $P_1 - P_0$ = $54 − $50	**$4.00**
2	Total dollar return = capital gain + dividend = $4.00 + $1.50	**$5.50**
3	HPR = $5.50 ÷ $50	**11.0%**
4	Split: capital-gain yield = $4 ÷ $50; dividend yield = $1.50 ÷ $50	**8.0% + 3.0% = 11.0%**

A realized HPR is historical and observable. For decisions we need the **expected return** E[r], the return we anticipate before the outcome is known. It can be estimated from the average of past returns or from a model such as the CAPM, which comes later in the course.

Risk is uncertainty about that return. We measure it with the **standard deviation** σ, the typical size of the deviations of returns around their mean:

$$\sigma = \sqrt{E[(r_t - \bar{r})^2]}$$

A higher σ means outcomes are more spread out and harder to predict. Because investors dislike that uncertainty, they require a higher expected return to hold riskier assets. The extra expected return over a risk-free rate is the **risk premium**.

WORKED EXAMPLE 3

Mean and standard deviation from five annual returns

A stock returned 12%, −8%, 20%, 5% and 16% over five years. (Illustrative numbers.)

Year	Return r_t	Deviation $r_t - \bar{r}$	Squared deviation
1	12%	12 − 9 = 3	9
2	−8%	−8 − 9 = −17	289
3	20%	20 − 9 = 11	121
4	5%	5 − 9 = −4	16
5	16%	16 − 9 = 7	49
Sum	45%	0 (always)	**484**

Step	What we compute	Result
1	Mean $\bar{r}$ = 45% ÷ 5	**9.0%**
2	Squared deviations (table above), summed	**484 (%²)**
3	Variance, dividing by n = 5 (the slide's E[·] definition)	**96.8 (%²)**
4	σ = √96.8	**9.84%**
5	Sample version, dividing by n − 1 = 4: variance = 121, σ = √121	**11.0%**

Dividing by n treats the five years as the entire population of outcomes. When we use a sample of history to *estimate* an unknown future σ, statistics packages and Excel's STDEV.S divide by n − 1, which gives a slightly larger number. With long histories the difference becomes negligible.

Key insight

We describe every asset by a pair of numbers, (E[r], σ), never by return alone. Two assets can offer the same expected return with very different risk. The goal is the most expected return per unit of risk.

4. Part 2: Modern Portfolio Theory

Modern Portfolio Theory (MPT), developed by Harry Markowitz, is a systematic way to find the best portfolio for an investor. Its **inputs** are the expected return and standard deviation of each security, the correlation between every pair of securities, and the investor's tolerance for risk. The **process** computes the expected return and risk of every possible combination of weights, identifies the efficient frontier, and ranks portfolios by their risk-return trade-off. The **output** is the optimal portfolio for that investor: the highest return per unit of risk they are willing to bear.

The central idea is that **a security should not be judged on its own**. What matters is how it changes the risk of the whole portfolio. A volatile stock that tends to move opposite to the rest of your holdings can *reduce* total portfolio risk.

The two-asset formulas

For a portfolio with weight w_A in asset A and $w_B = 1 - w_A$ in asset B:

$$E[r_p] = w_A \cdot E[r_A] + w_B \cdot E[r_B]$$

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2 \cdot w_A \cdot w_B \cdot \sigma_A \cdot \sigma_B \cdot \rho_{AB}$$

The expected return is a simple weighted average, and **correlation plays no role in it**. The variance has three terms: each asset's own variance scaled by its squared weight, plus a cross-term. The cross-term contains the covariance, $\text{Cov}(A, B) = \sigma_A\sigma_B\rho_{AB}$, so the lower the correlation ρ, the smaller the portfolio variance. That is the whole source of the diversification benefit.

Correlation ρ	Meaning	Diversification benefit
+1	Perfect positive: the assets always move together	None. σ_p is just the weighted average of the σ's
0	Uncorrelated: movements are unrelated	Substantial
−1	Perfect negative: always move in opposite directions	Maximum. Some mix has $\sigma_p = 0$

Real stocks typically have correlations of about 0.3 to 0.6 with one another: positive but well below 1. So combining them always reduces risk relative to the weighted average, even though it can never eliminate it.

WORKED EXAMPLE 4

Portfolio return and risk: 60% stocks, 40% bonds (slide Example 1)

Asset	E[r]	σ	Weight
A: Stocks	15%	30%	$w_A = 60\%$
B: Bonds	7%	12%	$w_B = 40\%$

Correlation $\rho_{AB} = 0.20$. Find $E[r_p]$, σ_p^2 and σ_p. Work in decimals to keep the variance units consistent.

Step	What we compute	Result
1	Expected return = 0.60 × 15% + 0.40 × 7% = 9.0% + 2.8%	**$E[r_p] = 11.8\%$**
2a	Stock term = $w_A^2\sigma_A^2 = (0.6)^2 \times (0.30)^2 = 0.36 \times 0.09$	**0.032400**
2b	Bond term = $w_B^2\sigma_B^2 = (0.4)^2 \times (0.12)^2 = 0.16 \times 0.0144$	**0.002304**
2c	Cross-term = 2 × 0.6 × 0.4 × 0.30 × 0.12 × 0.20 = 0.48 × 0.036 × 0.20	**0.003456**
2	Variance = 0.032400 + 0.002304 + 0.003456	**$\sigma_p^2 = 0.03816$**
3	Standard deviation = √0.03816	**$\sigma_p = 19.5\%$**
4	Compare: weighted-average σ = 0.6 × 30% + 0.4 × 12% = 18% + 4.8%	**22.8%**

The portfolio's risk (19.5%) is well below the weighted average of the two risks (22.8%), while its expected return is exactly the weighted average. The 3.3-point reduction in risk costs nothing in return. It comes entirely from the correlation being less than +1.

WORKED EXAMPLE 5

How much does correlation matter?

Keep the same assets and weights and change only ρ. Terms 2a and 2b do not depend on ρ and always sum to 0.034704. The cross-term is $2 \times 0.6 \times 0.4 \times 0.30 \times 0.12 \times \rho = 0.01728 \times \rho$.

Correlation ρ	Cross-term $0.01728 \times \rho$	Variance σ_p^2	σ_p	$E[r_p]$
+1 (no benefit)	+0.017280	0.034704 + 0.017280 = 0.051984	22.8%	11.8%
+0.20 (our case)	+0.003456	0.034704 + 0.003456 = 0.038160	19.5%	11.8%
0	0	0.034704	18.6%	11.8%
−1 (maximum)	−0.017280	0.034704 − 0.017280 = 0.017424	13.2%	11.8%

Two quick checks. At $\rho = +1$ the variance is a perfect square, so $\sigma_p = 0.6 \times 30\% + 0.4 \times 12\% = 22.8\%$, the weighted average. At $\rho = -1$ it is also a perfect square, so $\sigma_p = |0.6 \times 30\% - 0.4 \times 12\%| = |18\% - 4.8\%| = 13.2\%$. As ρ falls from +1 to −1, risk drops from 22.8% to 13.2% while the expected return stays at 11.8%. Same reward, less risk: the free lunch of diversification.

WORKED EXAMPLE 6

Two special portfolios: zero risk and minimum variance

Zero risk when ρ = −1. With perfectly negative correlation, the two risks can cancel exactly. Set $\sigma_p = |w_A\sigma_A - (1 - w_A)\sigma_B| = 0$ and solve:

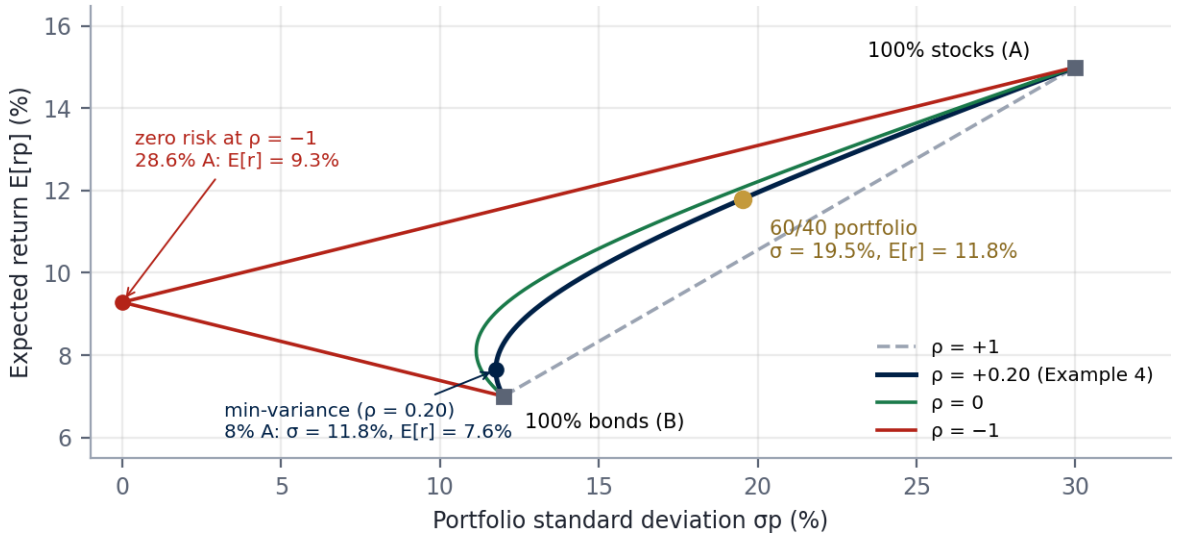
Step	What we compute	Result
1	$w_A \times 30\% = (1 - w_A) \times 12\%$	$\mathbf{30w_A = 12 - 12w_A}$
2	$w_A = \sigma_B \div (\sigma_A + \sigma_B) = 12 \div 42$	$\mathbf{w_A = 28.6\%, w_B = 71.4\%}$
3	$E[r_p] = 0.286 \times 15\% + 0.714 \times 7\%$	**9.29% with $\sigma_p = 0$**

Minimum variance when ρ = 0.20. With realistic correlation, risk cannot reach zero, but some mix has the lowest possible risk. Its weight on A is:

$$w_A^* = (\sigma_B^2 - \sigma_A\sigma_B\rho) \div (\sigma_A^2 + \sigma_B^2 - 2\sigma_A\sigma_B\rho)$$

Step	What we compute	Result
1	Numerator = 0.0144 − (0.30)(0.12)(0.20) = 0.0144 − 0.0072	**0.0072**
2	Denominator = 0.09 + 0.0144 − 2(0.0072) = 0.09	**0.0900**
3	$w_A^* = 0.0072 \div 0.09$	**8% stocks, 92% bonds**
4	$E[r_p] = 0.08 \times 15\% + 0.92 \times 7\%$	**7.64%**
5	$\sigma_p^2 = 0.08^2(0.09) + 0.92^2(0.0144) + 2(0.08)(0.92)(0.0072) = 0.000576 + 0.012188 + 0.001060$	**0.013824**
6	$\sigma_p = \sqrt{0.013824}$	**11.76%**

Compare this with holding 100% bonds: σ = 12% and E[r] = 7%. Moving just 8% of the money into the *riskier* asset lowers risk to 11.76% *and* raises the expected return to 7.64%. An all-bond portfolio is therefore **inefficient**, and no investor who cares only about E[r] and σ should hold it. Adding a volatile asset with low correlation made the portfolio safer.



Every mix of stocks (A) and bonds (B) for four correlations. Lower ρ bends the curve further left: the same expected returns at lower risk. Only the upper part of each curve, from the minimum-variance point upward, is efficient.

5. The efficient frontier

The interactive slide let you move the weight and the correlation and watch the curve change. For two assets, every possible portfolio lies on a single curve. The part of that curve that runs from the minimum-variance portfolio upward is the **efficient frontier**: for each level of risk, it offers the highest expected return. Portfolios on the lower part of the curve are inefficient, because another portfolio with the same σ offers more return.

The last slide built the frontier step by step with nine asset classes, from Treasury bonds (σ 11%, E[r] 4%) through corporate bonds, REITs, value, blend and growth stocks, and international developed markets, to small caps and emerging markets (σ 35%, E[r] 27%). With many assets, the possible portfolios fill a whole region rather than a single line, and the individual assets sit to the right of the frontier: each can be beaten by a diversified portfolio with the same return and less risk. The efficient frontier is the **upper-left boundary** of that region. When correlations are lower, the whole frontier shifts to the left and every investor gets a better risk-return trade-off.

6. Key takeaways

Return and risk come as a pair. A single asset's return is its capital gain plus income, and its risk is the standard deviation σ of that return. Investors require a risk premium for bearing higher σ, so assets are compared on (E[r], σ), not return alone.

Portfolio return is an average; portfolio risk is not. $E[r_p]$ is always the weighted average of the assets' expected returns and does not depend on correlation. σ_p^2 contains the cross-term $2w_A w_B \sigma_A \sigma_B \rho$, so portfolio risk falls as correlation falls.

Correlation drives the diversification benefit. At $\rho = +1$ there is no benefit. Below +1 the portfolio is less risky than the weighted average, and at $\rho = -1$ risk can be removed entirely. Real stock correlations of 0.3 to 0.6 mean diversification always helps.

The efficient frontier is where rational investors choose. It is the upper-left boundary of all possible portfolios: the maximum expected return for each level of risk. Lower correlations shift it left.

The free lunch of investing

Combining assets with low or negative correlation reduces portfolio risk without reducing expected return. Changing ρ leaves $E[r_p]$ unchanged but shrinks σ_p. This asymmetry is the central message of Modern Portfolio Theory. **Next lecture:** asset allocation, which combines the risky portfolio with a risk-free asset through the Capital Allocation Line (CAL) to find the optimal risky portfolio.

7. Practice problems

Work these out before reading the solutions. Each one uses a method from the examples above.

Problem 1. You buy a stock at \$80, receive a \$2 dividend, and sell it a year later at \$76. What is your holding-period return? Split it into its capital-gain and dividend components.

Problem 2. A fund returned 10%, −5%, 15% and 0% over four years. Compute its mean return and its standard deviation, both dividing by n and dividing by n − 1.

Problem 3. Asset X has $E[r] = 10\%$ and $\sigma = 20\%$; asset Y has $E[r] = 5\%$ and $\sigma = 10\%$. You invest 50% in each and the correlation is 0.5. Find $E[r_p]$ and σ_p. What would σ_p be if the correlation were −1?

Problem 4. Using X and Y from Problem 3 with $\rho = -1$, what weights give a zero-risk portfolio, and what is its expected return?

Problem 5. You short 200 shares at \$40 and cover at \$46. Ignoring fees, what is your profit or loss in dollars and as a percentage of the short-sale price? What is the most you could have gained?

Examples 1, 4 and 5 use the figures from the Lecture 03 slides; Examples 2, 3 and 6 use illustrative numbers.

Solutions

	Work	Answer
P1	HPR = ($76 − $80 + $2) ÷ $80 = −$2 ÷ $80	**−2.5%**
	Capital-gain yield = −$4 ÷ $80 = −5.0%; dividend yield = $2 ÷ $80 = +2.5%	**−5.0% + 2.5%**
P2	Mean = (10 − 5 + 15 + 0) ÷ 4 = 20 ÷ 4	**5.0%**
	Deviations 5, −10, 10, −5; squares 25 + 100 + 100 + 25	**250**
	Divide by n: √(250 ÷ 4) = √62.5	**σ = 7.91%**
	Divide by n − 1: √(250 ÷ 3) = √83.3	**σ = 9.13%**
P3	$E[r_p] = 0.5 \times 10\% + 0.5 \times 5\%$	**7.5%**
	$\sigma_p^2 = 0.25(0.04) + 0.25(0.01) + 2(0.5)(0.5)(0.20)(0.10)(0.5) = 0.0100 + 0.0025 + 0.0050$	**0.0175**
	$\sigma_p = \sqrt{0.0175}$	**13.23%**
	ρ = −1: $\sigma_p = \lvert 0.5 \times 20\% - 0.5 \times 10\% \rvert$	**5.0%**
P4	$w_X = \sigma_Y \div (\sigma_X + \sigma_Y) = 10 \div 30$	**33.3% X, 66.7% Y**
	$E[r_p] = (1/3)(10\%) + (2/3)(5\%) = 3.33\% + 3.33\%$	**6.67%, $\sigma_p = 0$**
P5	Per share: $40 − $46 = −$6; total −$6 × 200	**−$1,200**
	Percentage = −$6 ÷ $40	**−15.0%**
	Maximum gain if the price falls to $0: $40 × 200	**$8,000 (100%)**